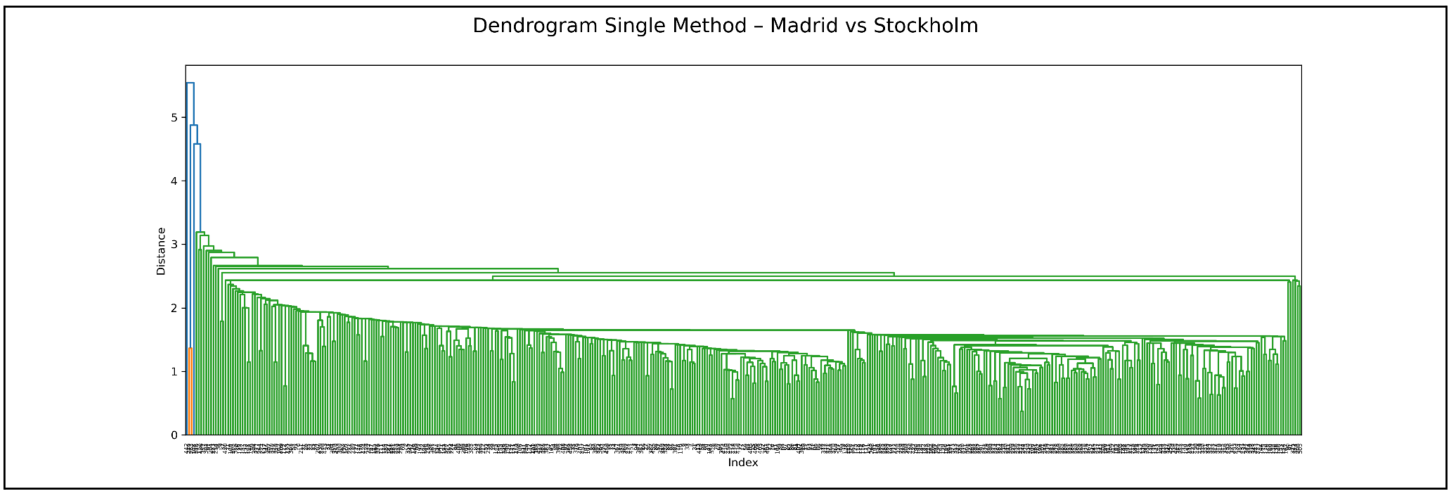
CareerFoundry ML with Python
December 16, 2025
Charlotte Lin

Achievement 2.1: Unsupervised Learning Algorithms

Dendrograms

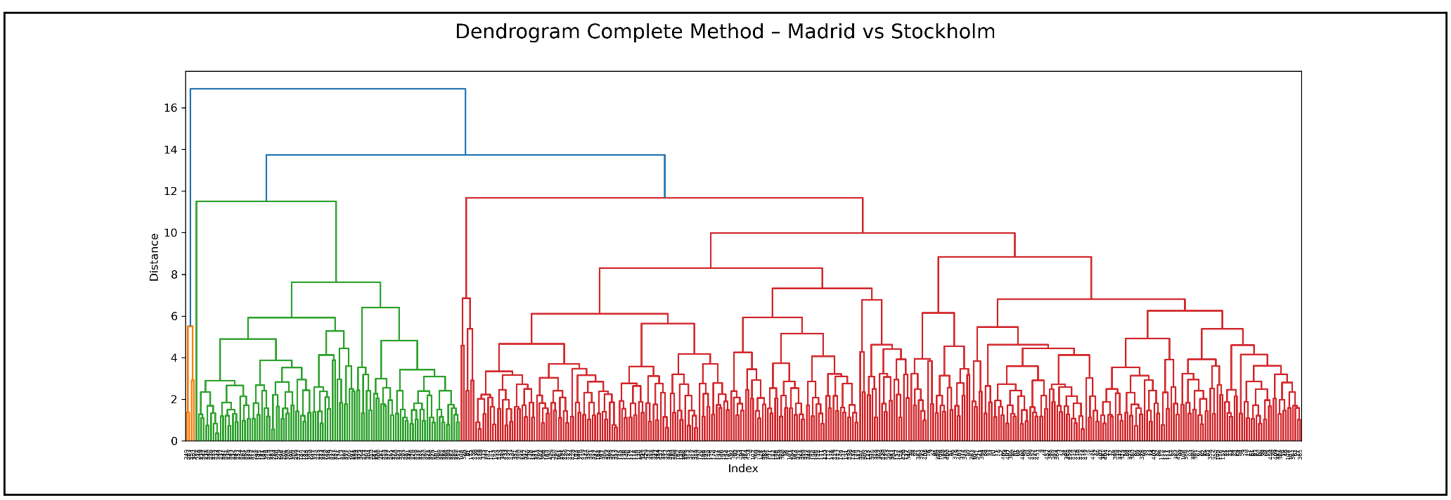
Madrid vs Stockholm (2010)

Single Method



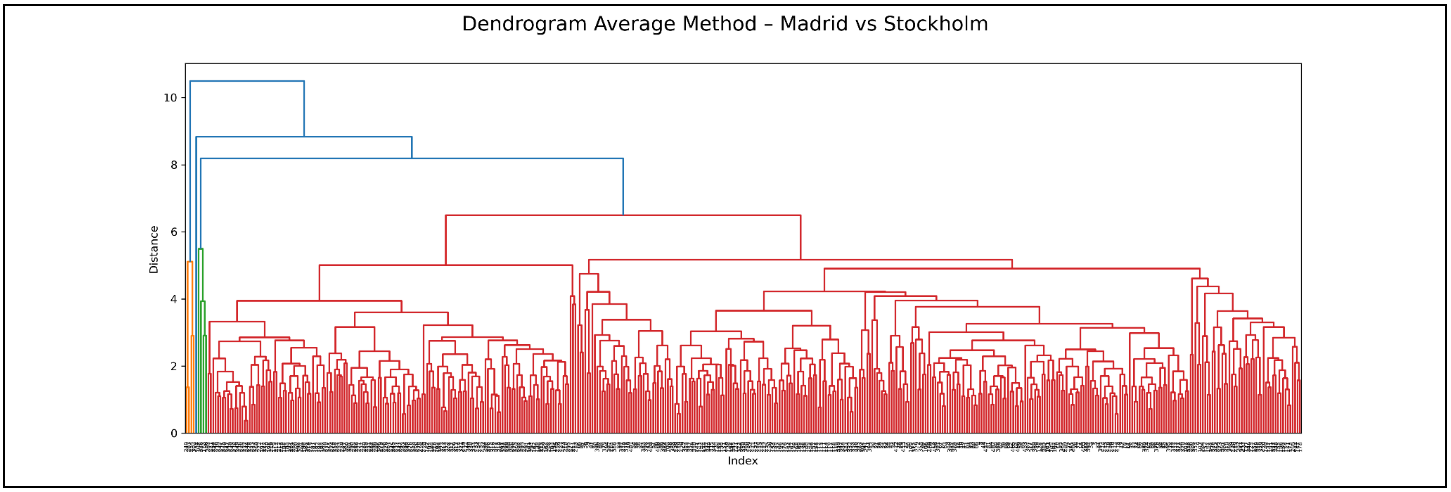
The single-method dendrogram exhibits a strong chaining effect that does not form distinct clusters. This outcome suggests it is unsuitable for identifying coherent weather patterns in the ClimateWins dataset.

Complete Method



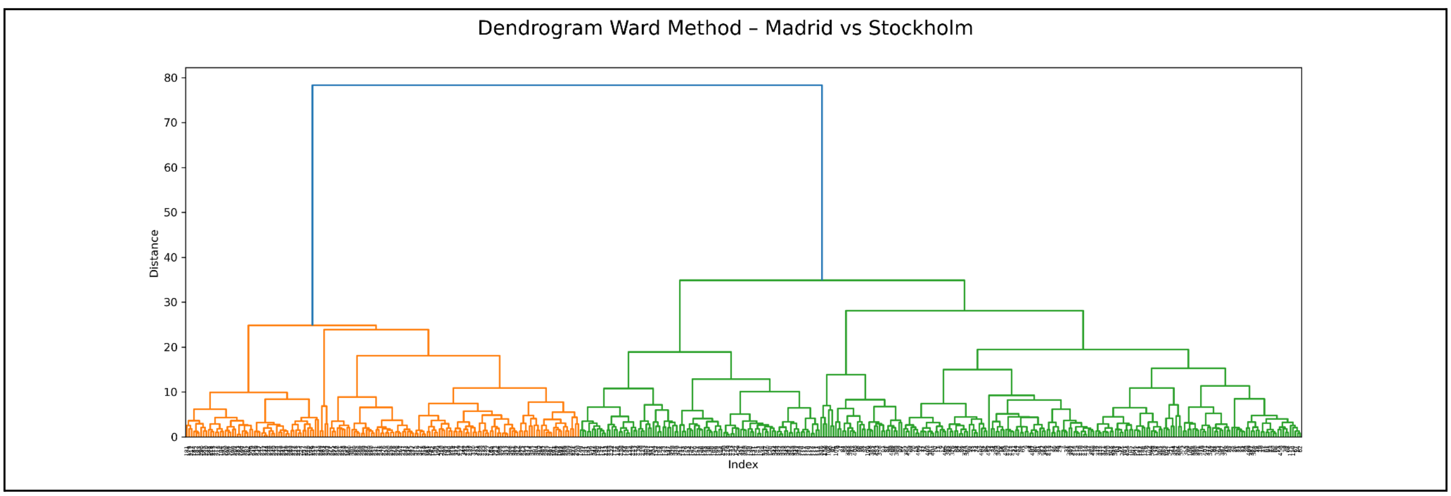
The complete-method dendrogram shows clearer separation and more compact clusters than the single-method. The clusters reveal structural differences in weather patterns between Madrid and Stockholm, likely reflecting underlying seasonal and climatic differences.

Average Method



The average-method dendrogram displays a more balanced clustering structure. The pattern likely reflects seasonal transitions and partial overlap in weather conditions between Madrid and Stockholm.

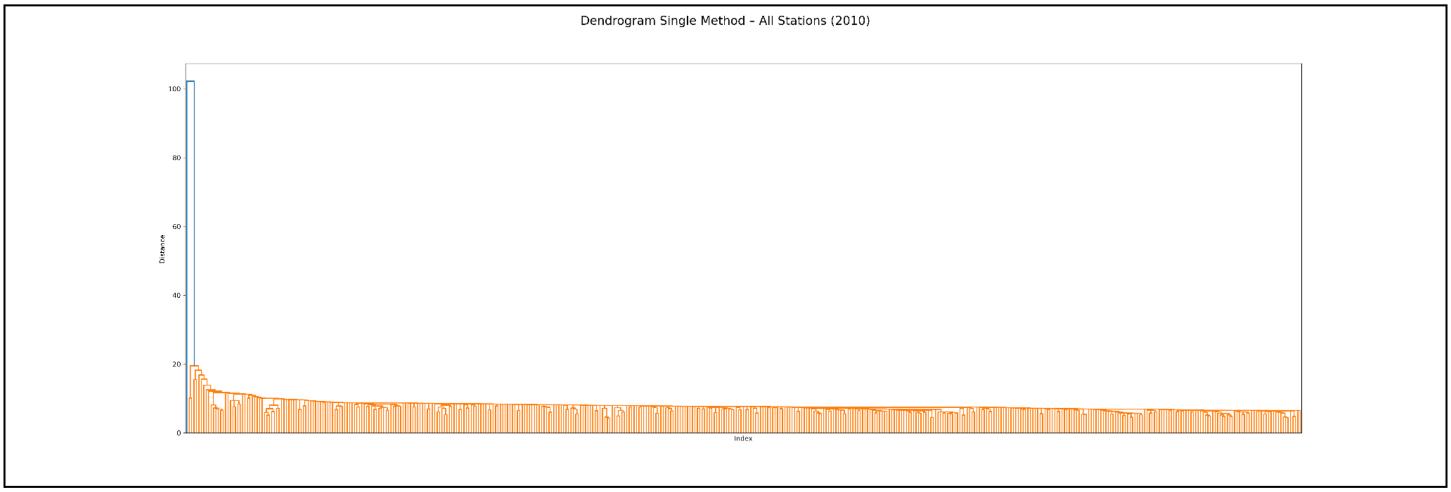
Ward Method



The Ward-method dendrogram shows the clearest high-level separation among the four linkage methods. A large jump in linkage distance at the final merge indicates that the daily observations naturally partition into two distinct macro-clusters, which suggests strongly differentiated weather regimes within the 2010 data. Compared to single, complete, and average methods, Ward produces more compact and interpretable cluster structure because it minimizes within-cluster variance. This makes Ward linkage the strongest candidate for downstream validation against ClimateWins' "pleasant weather" labels.

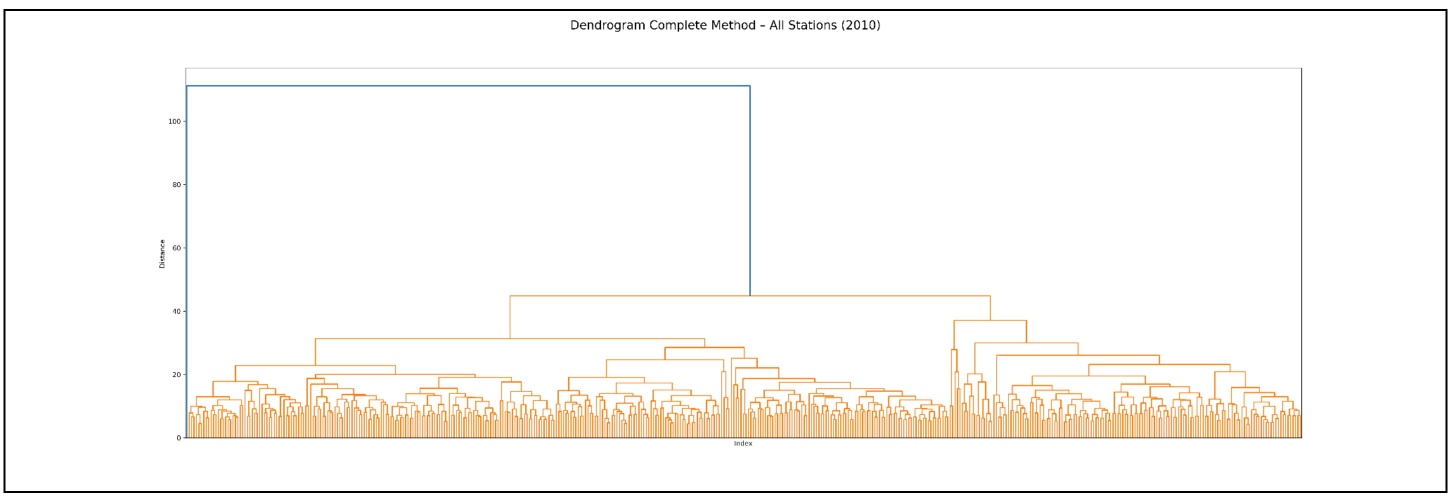
All Cities (2010)

Single Method



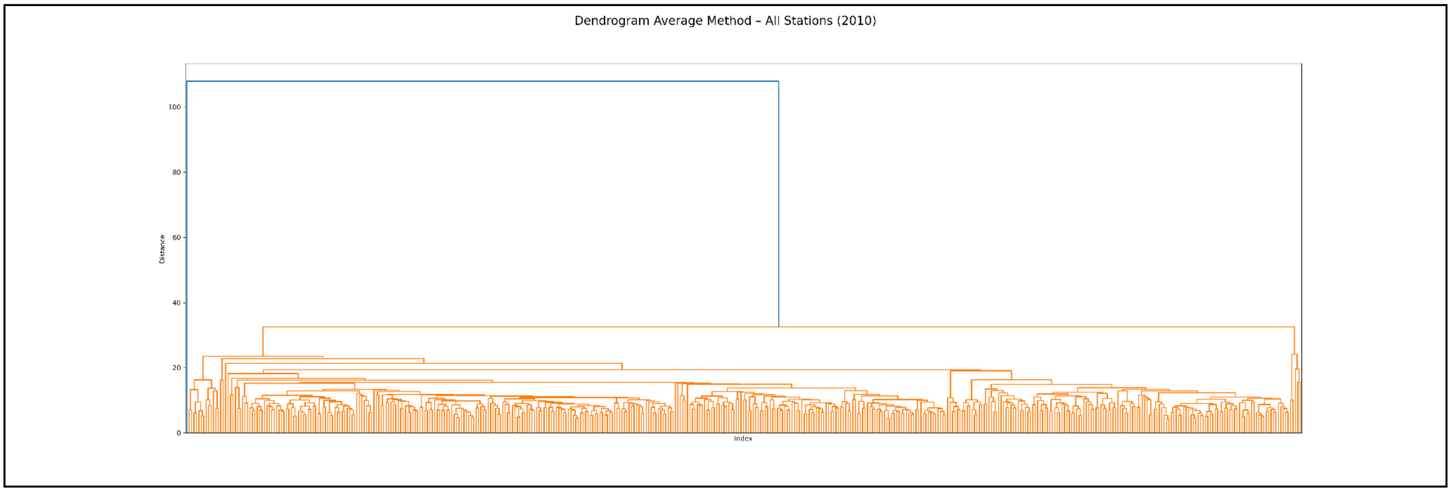
The single-method dendrogram produces a dense "comb" structure rather than distinct, compact clusters. This pattern suggests that it is not suited for identifying weather regimes across all stations in the ClimateWins dataset.

Complete Method



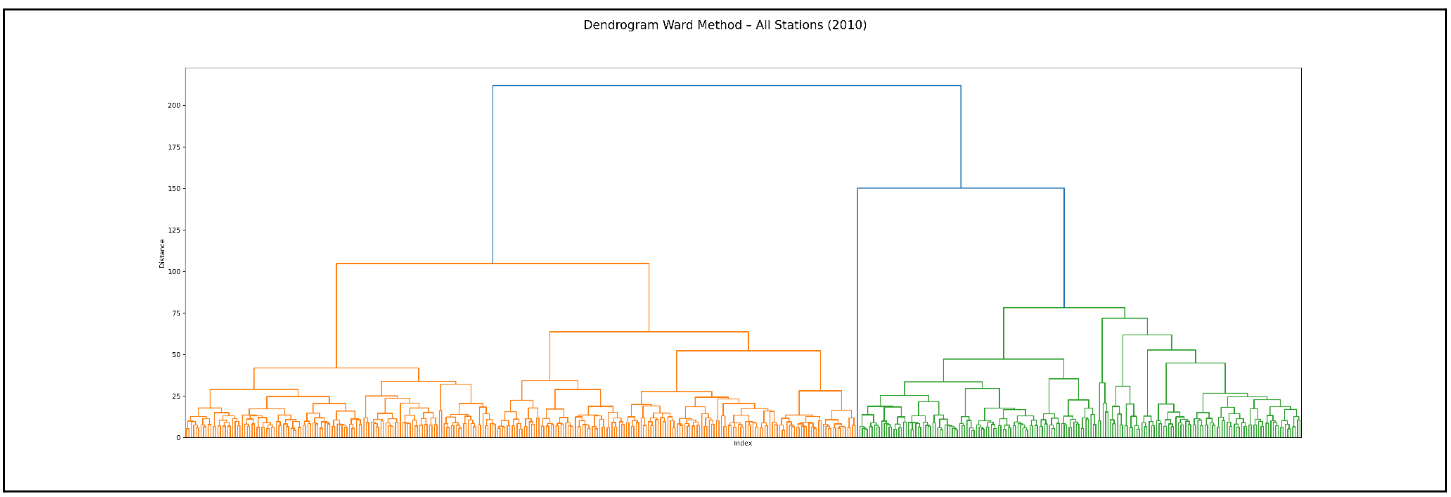
The complete method dendrogram reveals two dominant clusters merging only at a high linkage distance. Compared to single linkage, complete linkage produces more compact and balanced clusters, with much better interpretability. The structure suggests that daily weather observations across all stations group into a small number of broad climatic regimes, with meaningful substructure within each regime.

Average Method



The average-method dendrogram shows a clear high-level separation of daily weather observations with two dominant clusters. Compared to complete linkage, cluster merging occurs more gradually with smoother transitions between weather regimes.

Ward Method



The Ward-method dendrogram reveals the strongest and most interpretable cluster structure among all linkage methods. A very large increase in linkage distance at the final merge indicates that daily weather observations across all stations naturally divide into two highly distinct macro-clusters. Within each cluster, the hierarchical structure is compact and well balanced. This result suggests that Ward linkage is the most suitable method for identifying meaningful weather regimes in the ClimateWins dataset.

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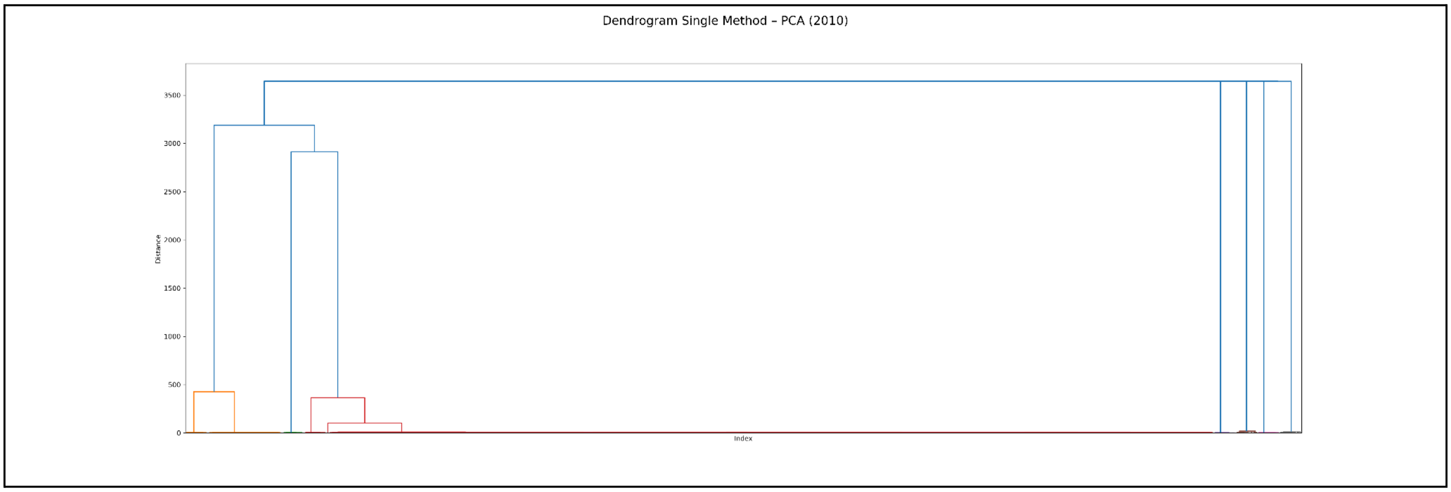
Dendrograms with PCA

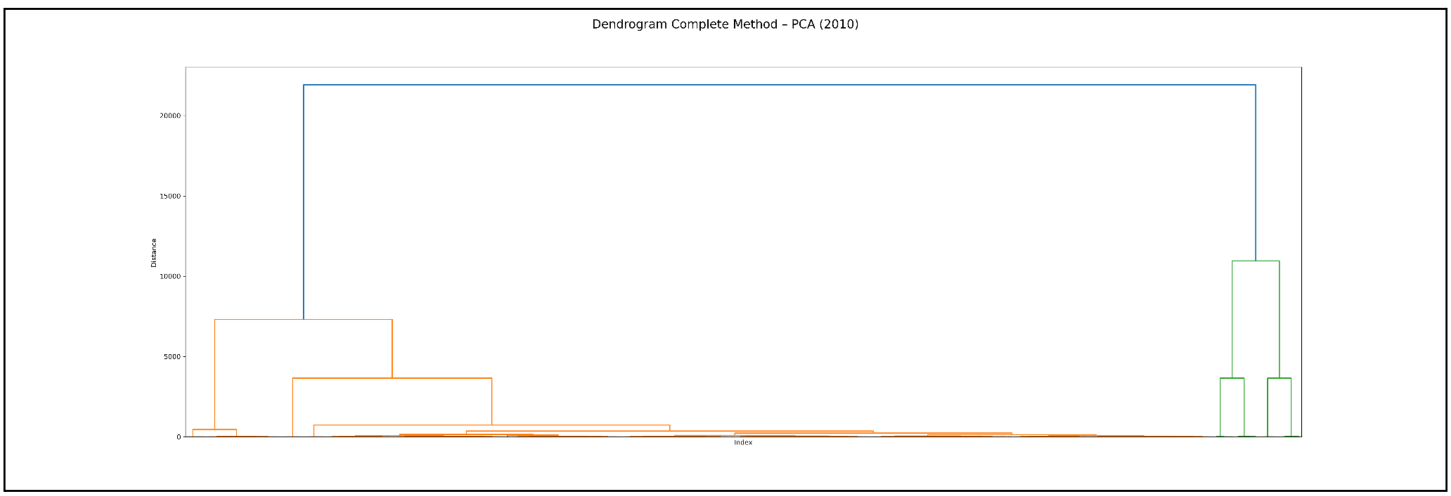
All Cities (2010)

Single Method



The single-linkage dendrogram in PCA space exhibits a pronounced chaining effect. This structure indicates that single linkage does not produce clearly separable clusters even in the reduced PCA space.

Complete Method



The complete-linkage dendrogram in PCA space reveals a clear high-level separation, with clusters merging only at large linkage distances. Compared to single linkage, complete linkage produces more compact and well-separated clusters after dimensionality reduction, which indicates that PCA enhances the interpretability of cluster structure under this method.

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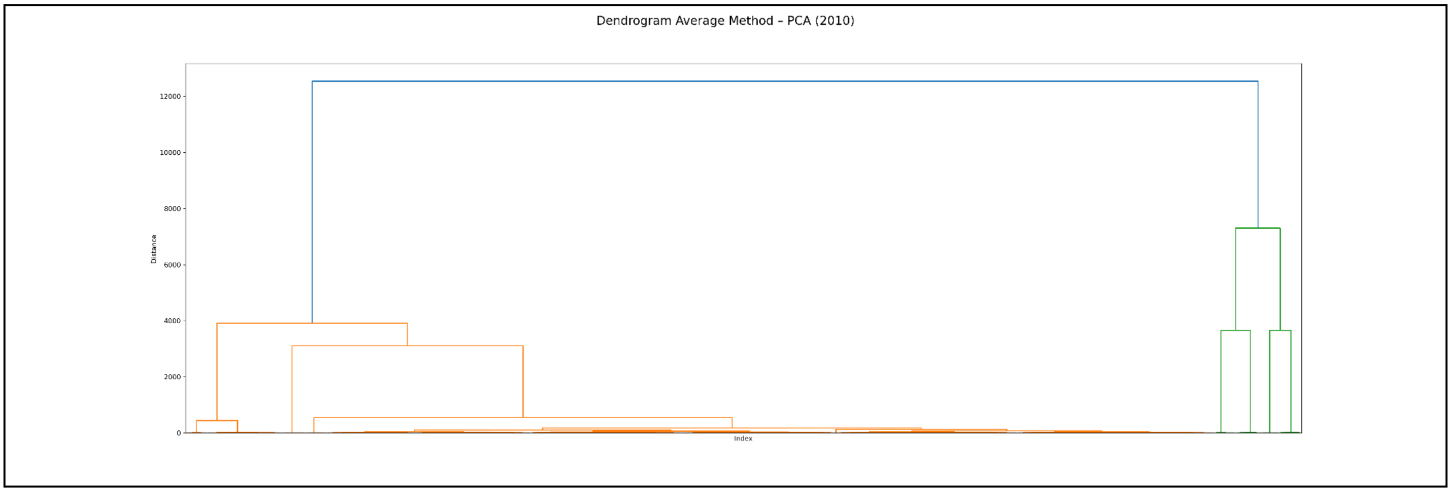
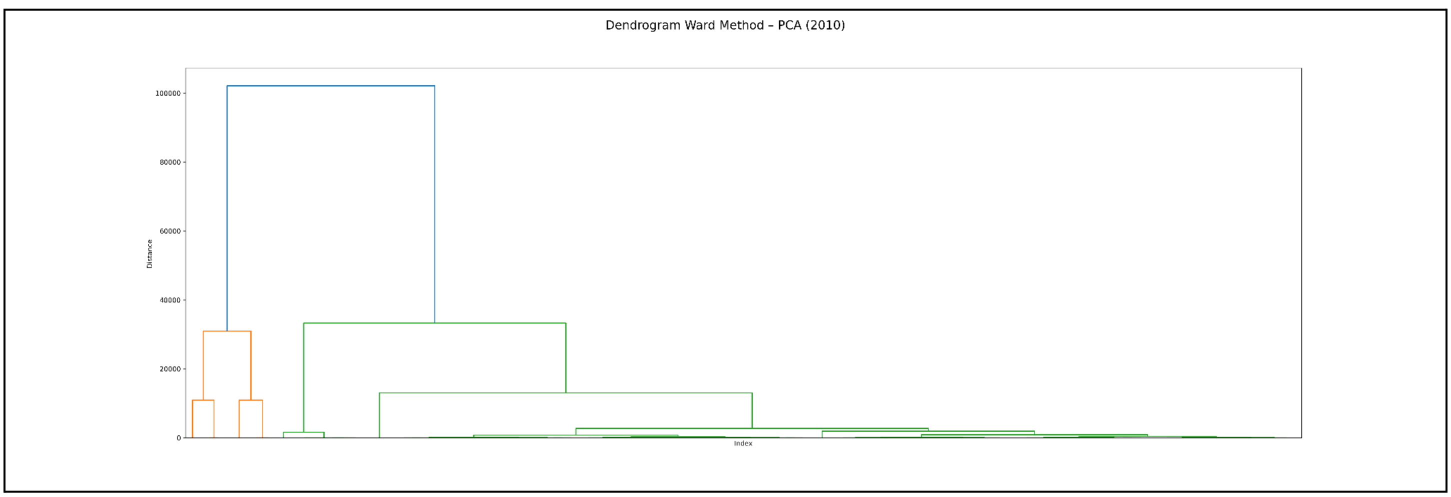
Average Method



The average-linkage dendrogram in PCA space displays a clear macro-level separation. This pattern suggests that average linkage captures broad weather regimes while allowing smoother transitions between clusters. This reflects overlapping climatic conditions even after dimensionality reduction.

Ward Method



The Ward-linkage dendrogram in PCA space exhibits the most distinct and interpretable clustering structure, with clear hierarchical separation and large jumps in linkage distance at higher levels. This indicates that Ward's method, combined with PCA, effectively groups observations by minimizing within-cluster variance and reveals dominant weather pattern regimes in the reduced-dimensional space.